

MongoDB

```
}
}
test> db.books.countDocuments()
30
test> db.books.insertMany([
|   { title: "MongoDB in Action", author: "Kyle Banker", category: "Databases", price: 45, in_stock: true, published_year: 2016, rating: 4.4 },
|   { title: "Redis in Action", author: "Josiah L. Carlson", category: "Databases", price: 47, in_stock: true, published_year: 2013, rating: 4.5 },
|   { title: "Database Internals", author: "Alex Petrov", category: "Databases", price: 60, in_stock: false, published_year: 2019, rating: 4.7 },
|   { title: "Learning SQL", author: "Alan Beaulieu", category: "Databases", price: 40, in_stock: true, published_year: 2020, rating: 4.3 },
|   { title: "SQL Antipatterns", author: "Bill Karwin", category: "Databases", price: 44, in_stock: true, published_year: 2010, rating: 4.2 }
| ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a4ce2a114289789f6208a80'),
    '1': ObjectId('6a4ce2a114289789f6208a81'),
    '2': ObjectId('6a4ce2a114289789f6208a82'),
    '3': ObjectId('6a4ce2a114289789f6208a83'),
    '4': ObjectId('6a4ce2a114289789f6208a84')
  }
}
test> db.books.countDocuments()
35
test>
```

```
test> db.books.find({ category: "Programming" })
[
  {
    _id: ObjectId('6a4ce29114289789f6208a62'),
    title: 'Clean Code',
    author: 'Robert C. Martin',
    category: 'Programming',
    price: 35,
    in_stock: true,
    published_year: 2008,
    rating: 4.8
  },
  {
    _id: ObjectId('6a4ce29114289789f6208a63'),
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    category: 'Programming',
    price: 42,
    in_stock: true,
    published_year: 1999,
    rating: 4.7
  },
  {
    _id: ObjectId('6a4ce29114289789f6208a64'),
    title: 'Design Patterns',
    author: 'Erich Gamma',
    category: 'Programming',
    price: 50,
    in_stock: true,
```

```
test> db.books.find({ published_year: { $gt: 2015 } })
[
  {
    _id: ObjectId('6a4ce29114289789f6208a67'),
    title: 'Python Crash Course',
    author: 'Eric Matthes',
    category: 'Programming',
    price: 39,
    in_stock: true,
    published_year: 2019,
    rating: 4.7
  },
  {
    _id: ObjectId('6a4ce29114289789f6208a68'),
    title: 'Fluent Python',
    author: 'Luciano Ramalho',
    category: 'Programming',
    price: 55,
    in_stock: true,
    published_year: 2022,
    rating: 4.8
  },
  {
    _id: ObjectId('6a4ce29114289789f6208a69'),
    title: 'Effective Java',
    author: 'Joshua Bloch',
    category: 'Programming',
    price: 48,
    in_stock: false,
    published_year: 2018,
```

```
]
test> db.books.find({ price: { $gt: 40 } })
[
  {
    _id: ObjectId('6a4ce29114289789f6208a63'),
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    category: 'Programming',
    price: 42,
    in_stock: true,
    published_year: 1999,
    rating: 4.7
  },
  {
    _id: ObjectId('6a4ce29114289789f6208a64'),
    title: 'Design Patterns',
    author: 'Erich Gamma',
    category: 'Programming',
    price: 50,
    in_stock: true,
    published_year: 1994,
    rating: 4.6
  },
  {
    _id: ObjectId('6a4ce29114289789f6208a65'),
    title: 'Introduction to Algorithms',
    author: 'Thomas H. Cormen',
```

```

test> db.books.find({ in_stock: true })
[
  {
    _id: ObjectId('6a4ce29114289789f6208a62'),
    title: 'Clean Code',
    author: 'Robert C. Martin',
    category: 'Programming',
    price: 35,
    in_stock: true,
    published_year: 2008,
    rating: 4.8
  },
  {
    _id: ObjectId('6a4ce29114289789f6208a63'),
    title: 'The Pragmatic Programmer',
    author: 'Andrew Hunt',
    category: 'Programming',
    price: 42,
    in_stock: true,
    published_year: 1999,
    rating: 4.7
  },
  {
    _id: ObjectId('6a4ce29114289789f6208a64'),
    title: 'Design Patterns',
    author: 'Erich Gamma',
    category: 'Programming',
    price: 50,
    in_stock: true,
    published_year: 1994,
  }
]

```

```

Type "it" for more
test> db.books.find({ author: "Robert C. Martin" })
[
  {
    _id: ObjectId('6a4ce29114289789f6208a62'),
    title: 'Clean Code',
    author: 'Robert C. Martin',
    category: 'Programming',
    price: 35,
    in_stock: true,
    published_year: 2008,
    rating: 4.8
  }
]
test> |

```

```

test> db.books.updateOne(
|   { title: "Clean Code" },
|   { $set: { price: 38 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test> db.books.find({ title: "Clean Code" })
[
  {
    _id: ObjectId('6a4ce29114289789f6208a62'),
    title: 'Clean Code',
    author: 'Robert C. Martin',
    category: 'Programming',
    price: 38,
    in_stock: true,
    published_year: 2008,
    rating: 4.8
  }
]
test> |

```

```

]
test> db.books.updateOne(
|   { title: "Database Internals" },
|   { $set: { in_stock: true } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test> db.books.find({ title: "Database Internals" })
[
  {
    _id: ObjectId('6a4ce2a114289789f6208a82'),
    title: 'Database Internals',
    author: 'Alex Petrov',
    category: 'Databases',
    price: 60,
    in_stock: true,
    published_year: 2019,
    rating: 4.7
  }
]
test>

```

```

test> db.books.updateOne(
|   { title: "Python Crash Course" },
|   { $inc: { rating: 0.1 } }
| )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
test> db.books.find({ title: "Python Crash Course" })
[
  {
    _id: ObjectId('6a4ce29114289789f6208a67'),
    title: 'Python Crash Course',
    author: 'Eric Matthes',
    category: 'Programming',
    price: 39,
    in_stock: true,
    published_year: 2019,
    rating: 4.8
  }
]
test> |

```

```

test> db.books.deleteOne({ title: "Brave New World" })
{ acknowledged: true, deletedCount: 1 }
test> db.books.deleteOne({ title: "Homo Deus" })
{ acknowledged: true, deletedCount: 1 }
test> db.books.countDocuments()
33
test> db.books.find({ title: { $in: ["Brave New World", "Homo Deus"] } })
test>

```

```

test> db.books.aggregate([
|   {
|     $group: {
|       _id: "$category",
|       average_price: { $avg: "$price" }
|     }
|   }
| ])
[
  { _id: 'Business', average_price: 31.8 },
  { _id: 'Databases', average_price: 47.2 },
  { _id: 'Self-Development', average_price: 24.333333333333332 },
  { _id: 'Fiction', average_price: 18.8 },
  { _id: 'History', average_price: 32.25 },
  { _id: 'Programming', average_price: 46.875 }
]
test> |

```

```

test> db.books.aggregate([
|   {
|     $group: {
|       _id: "$category",
|       number_of_books: { $sum: 1 }
|     }
|   }
| ])
[
  { _id: 'Business', number_of_books: 5 },
  { _id: 'Databases', number_of_books: 5 },
  { _id: 'Self-Development', number_of_books: 6 },
  { _id: 'Fiction', number_of_books: 5 },
  { _id: 'History', number_of_books: 4 },
  { _id: 'Programming', number_of_books: 8 }
]
test> |

```



```

test> db.books.aggregate([
|   {
|     $group: {
|       _id: "$category",
|       average_rating: { $avg: "$rating" }
|     }
|   }
| ])
[
|   { _id: 'Business', average_rating: 4.46 },
|   { _id: 'Databases', average_rating: 4.42 },
|   { _id: 'Self-Development', average_rating: 4.516666666666667 },
|   { _id: 'Fiction', average_rating: 4.62 },
|   { _id: 'History', average_rating: 4.325 },
|   { _id: 'Programming', average_rating: 4.6375 }
| ]
test>

```

```

test> db.books.aggregate([
|   { $sort: { price: -1 } },
|   { $limit: 5 },
|   {
|     $project: {
|       _id: 0,
|       title: 1,
|       author: 1,
|       category: 1,
|       price: 1
|     }
|   }
| ])
[
|   {
|     title: 'Introduction to Algorithms',
|     author: 'Thomas H. Cormen',
|     category: 'Programming',
|     price: 75
|   },
|   {
|     title: 'Database Internals',
|     author: 'Alex Petrov',
|     category: 'Databases',
|     price: 60
|   },
|   {
|     title: 'Fluent Python',
|     author: 'Luciano Ramalho',
|     category: 'Programming',
|     price: 55
|   }
| ]

```

```

bookstore> db.books.find({
|   category: "Programming",
|   published_year: { $gte: 2020 }
| }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookstore.books',
    indexFilterSet: false,
    parsedQuery: {
      '$and': [
        { category: { '$eq': 'Programming' } } },
        { published_year: { '$gte': 2020 } } }
      ]
    },
    queryHash: 'FD1D2CA5',
    planCacheKey: 'FD1D2CA5',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    winningPlan: {
      stage: 'COLLSCAN',
      filter: {
        '$and': [
          { category: { '$eq': 'Programming' } } },
          { published_year: { '$gte': 2020 } } }
        ]
      },
      direction: 'forward'
    },
    rejectedPlans: []
  }
}

```

```

bookstore> db.books.createIndex({
|   category: 1,
|   published_year: 1
| })
category_1_published_year_1
bookstore> db.books.find({
|   category: "Programming",
|   published_year: { $gte: 2020 }
| }).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'bookstore.books',
    indexFilterSet: false,
    parsedQuery: {
      '$and': [
        { category: { '$eq': 'Programming' } },
        { published_year: { '$gte': 2020 } }
      ]
    },
    queryHash: 'FD1D2CA5',
    planCacheKey: '07DBF257',
    optimizationTimeMillis: 1,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    winningPlan: {
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { category: 1, published_year: 1 },

```

REDIS PART

```

127.0.0.1:6379> docker exec -it redis-assignment redis-cli
C:\Users\asti>docker exec -it redis-assignment redis-cli
127.0.0.1:6379> SET product:1:name "Wireless Mouse"
OK
127.0.0.1:6379> SET product:1:price "25.99"
OK
127.0.0.1:6379> SET product:1:category "Electronics"
OK
127.0.0.1:6379> GET product:1:name
"Wireless Mouse"
127.0.0.1:6379> GET product:1:price
"25.99"
127.0.0.1:6379> GET product:1:category
"Electronics"
127.0.0.1:6379>

```



```
127.0.0.1:6379> HSET product:2 name "Mechanical Keyboard" price "89.99" category "Electronics" stock "45"  
(integer) 4  
127.0.0.1:6379> HGETALL product:2  
1) "name"  
2) "Mechanical Keyboard"  
3) "price"  
4) "89.99"  
5) "category"  
6) "Electronics"  
7) "stock"  
8) "45"  
127.0.0.1:6379>
```

```
127.0.0.1:6379> RPUSH recent_orders order:1001 order:1002 order:1003 order:1004 order:1005 order:1006 order:1007 order:1008 order:1009 order:1010  
(integer) 10  
127.0.0.1:6379> LRange recent_orders 0 -1  
1) "order:1001"  
2) "order:1002"  
3) "order:1003"  
4) "order:1004"  
5) "order:1005"  
6) "order:1006"  
7) "order:1007"  
8) "order:1008"  
9) "order:1009"  
10) "order:1010"  
127.0.0.1:6379> LRange recent_orders -3 -1  
1) "order:1008"  
2) "order:1009"  
3) "order:1010"  
127.0.0.1:6379> |
```

```
(integer) 0  
127.0.0.1:6379> SADD product_tags wireless  
(integer) 0  
127.0.0.1:6379> SMEMBERS product_tags  
1) "electronics"  
2) "laptop"  
3) "accessories"  
4) "gaming"  
5) "wireless"  
6) "keyboard"  
7) "mouse"  
8) "office"  
127.0.0.1:6379> SISMEMBER product_tags gaming  
(integer) 1  
127.0.0.1:6379> SISMEMBER product_tags kitchen  
(integer) 0  
127.0.0.1:6379>
```

```
127.0.0.1:6379> EXPIRE featured_product 60
(integer) 1
127.0.0.1:6379> TTL featured_product
(integer) 53
127.0.0.1:6379> TTL featured_product
(integer) 43
127.0.0.1:6379> GET featured_product
"product:1"
127.0.0.1:6379> TTL featured_product
(integer) -2
127.0.0.1:6379> GET featured_product
(nil)
127.0.0.1:6379> |
```

```
127.0.0.1:6379> SET cache:product:1001 "Product page: Wireless Mouse, price 25.99, category Electronics" EX 120
OK
127.0.0.1:6379> GET cache:product:1001
"Product page: Wireless Mouse, price 25.99, category Electronics"
127.0.0.1:6379> TTL cache:product:1001
(integer) 98
127.0.0.1:6379> |
```

```
127.0.0.1:6379> ZADD sales_leaderboard 150 product:1 220 product:2 95 product:3 310 product:4 180 product:5
(integer) 5
127.0.0.1:6379> ZREVRANGE sales_leaderboard 0 -1 WITHSCORES
1) "product:4"
2) "310"
3) "product:2"
4) "220"
5) "product:5"
6) "180"
7) "product:1"
8) "150"
9) "product:3"
10) "95"
127.0.0.1:6379> ZREVRANGE sales_leaderboard 0 0 WITHSCORES
1) "product:4"
2) "310"
127.0.0.1:6379>
```